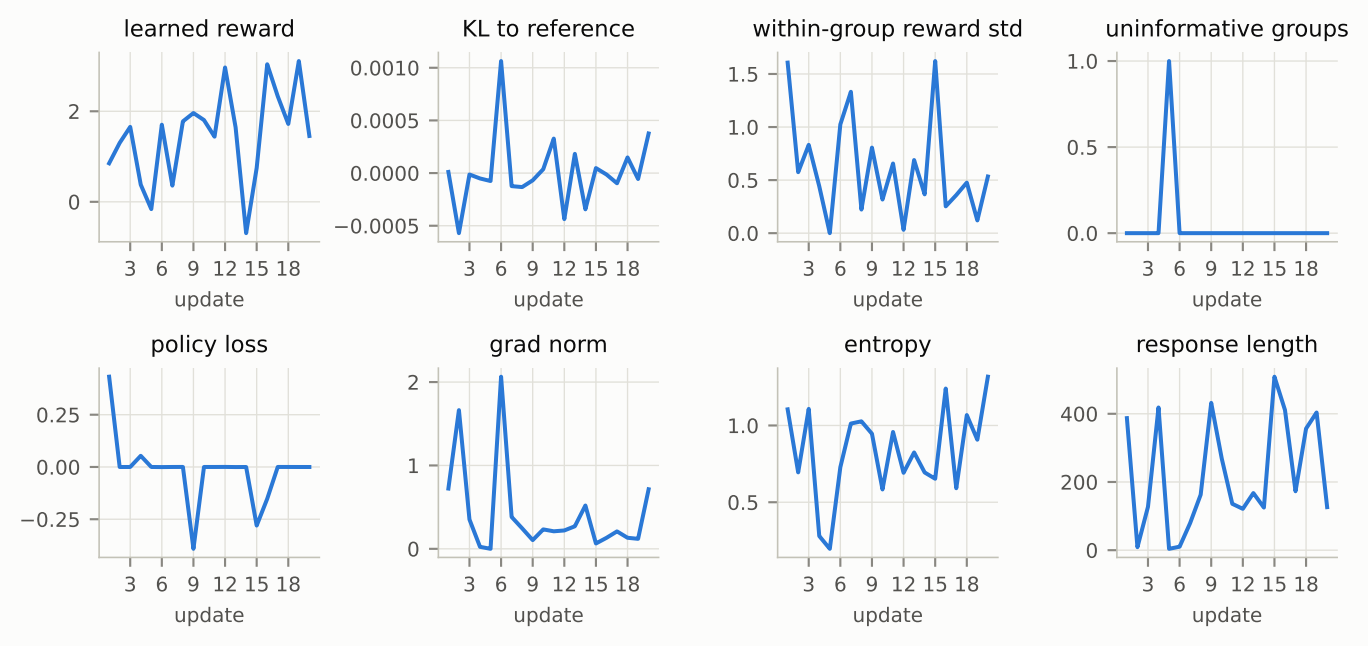
learned reward
2
0
3 6 9 12 15 18
update
KL to reference
0.0010
0.0005
0.0000
−0.0005
3 6 9 12 15 18
update
within-group reward std
1.5
1.0
0.5
0.0
3 6 9 12 15 18
update
uninformative groups
1.0
0.5
0.0
3 6 9 12 15 18
update
policy loss
0.25
0.00
−0.25
3 6 9 12 15 18
update
grad norm
2
1
0
3 6 9 12 15 18
update
entropy
1.0
0.5
3 6 9 12 15 18
update
response length
400
200
0
3 6 9 12 15 18
update